

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: T.Y. B. Tech. Branch: Mechanical Semester: V Subject Code & Name: BTMEC503 (Machine Design I) Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		
A)	Define and explain Fatigue failure, Endurance limit, Notch sensitivity	CO2	6
B)	Explain the Rankin's theory of failure	CO2	6
C)	Explain Maximum shear stress theory	CO3	6
Q.2	Solve Any Two of the following.		
A)	Assume the data of the transmission shaft given in Example 9.5. For this shaft, the permissible angle of twist is 3°/meter length. The modulus of rigidity of the shaft material is 79300 N/mm ² . Calculate: (i) the permissible angle of twist; and (ii) the shaft diameter on the basis of torsional rigidity.	CO2	6
B)	A solid shaft is transmitting 1 MW at 240 rpm. Determine the diameter of the shaft if the maximum torque transmitted exceeds the mean torque by 20%. Take the maximum allowable shear stress as 60 MPa	CO1	6
C)	Explain Aesthetic and ergonomic consideration in design	CO3	6
Q. 3	Solve Any One of the following.		
A)	Explain with diagram design procedure of knuckle joint	CO3	12
B)	What are the different types of springs? Explain with neat sketch.	CO2	12
C)	Explain Design procedure for Muff Coupling.	CO2	12
Q.4	Solve Any Two of the following.		
A)	Illustrate design procedure for square key	CO1	6
B)	What is a butt weld joint? Discuss design of butt weld joint under tension.	CO2	6
C)	Write note on Creep.	CO1	6

Q. 5	Solve Any One of the following.		
A)	It is required to design a helical compression spring subjected to a max. Force of 7.5 kN. The mean coil diameter should be 150 mm from space consideration. The spring rate is 75N/mm. The spring is made up of oil hardened and tempered steel with ultimate tensile strength of 1250 N/mm ² . The permissible shear stress for the spring wire is 30% of the ultimate tensile strength ($G = 81370 \text{ N/mm}^2$) Calculate: (1) Wire Diameter (2) No. of active coil.	CO3	12
B)	A rigid Coupling is used to transmit 50 kw power at 300 rpm. There are 6 bolts. The outer diameter of the flanges is 200 mm, while the recess diameter is 150 mm. The bolts are made of steel 45C8 ($S_{yt} = 380 \text{ N/mm}^2$) and the factor of safety is 3. Determine the diameter of the bolts. Assume the bolts are fitted in large clearance hole.	CO2	12
	*** End ***		

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